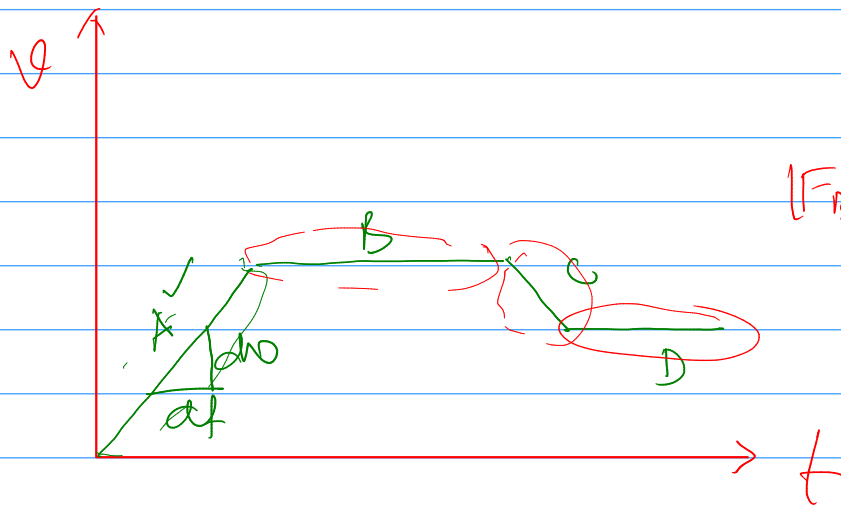


Newton's law

Force:

long range or contact force



$$|F_B| = |F_D| = 0$$

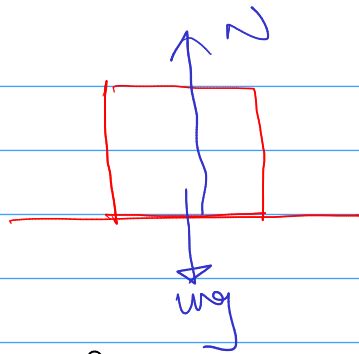
$$F_C < 0$$

$$|F_A| > 0 \quad |F_C| > 0$$

$$F = ma$$

Four Fundamental Forces:

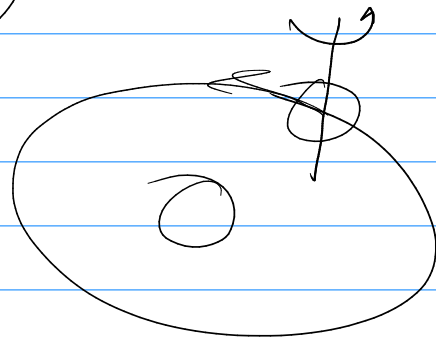
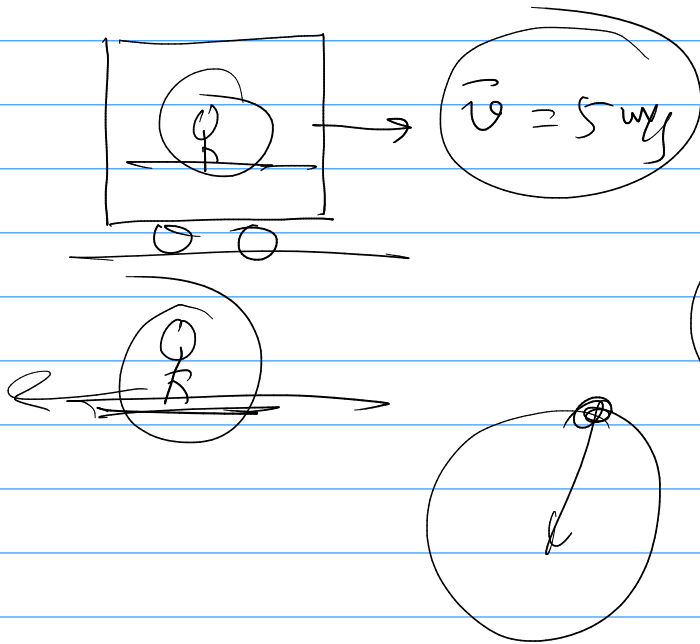
- (i) Strong force
 - (ii) Weak force
 - (iii) Gravitational force
 - (iv) Electromagnetic force
- Short range $\approx 10^{-18} \text{ m}$
- Long range $\approx \infty$



$$\vec{v}_{20}$$

Newton's law of Motion:-

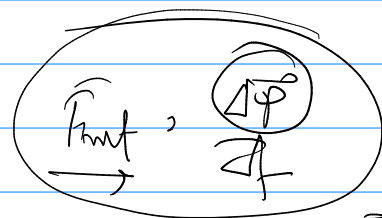
① Law of Inertia:-



$$a = 0.039 \text{ m/s}^2 \checkmark$$

Law II:-

$$\vec{F}_{\text{net}} = m\vec{a}$$



$$\vec{p} = m\vec{v}$$

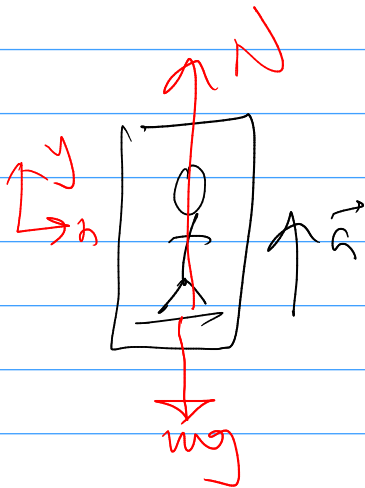
$$\frac{d\vec{p}}{dt} = m \frac{d\vec{v}}{dt} = m\vec{a} = \vec{F}$$

$$\vec{F}_{\text{net}} = \frac{d\vec{p}}{dt}$$

$$\vec{F} = m\vec{a}$$

$$\vec{p} = m\vec{v}$$

$$\frac{d\vec{p}}{dt} = m \frac{d\vec{v}}{dt} = m\vec{a} = \vec{F}_{\text{net}}$$



Free body diagram

$$\sum F_y = m\vec{a}$$

$$\Rightarrow N - mg = ma$$

$$\Rightarrow N = mg + ma$$

$$\Rightarrow \boxed{N = m(g + a)}$$

$$\sum F_y = m\vec{a}$$

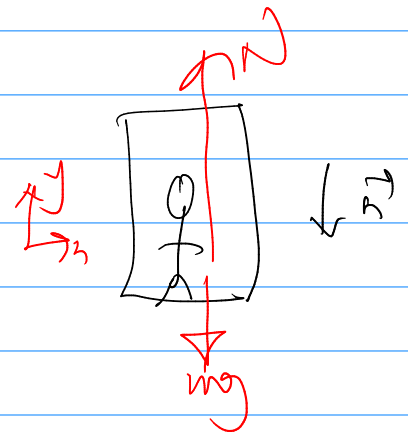
$$\Rightarrow N - mg = -ma$$

$$\Rightarrow N = mg - ma$$

$$\Rightarrow \boxed{N = m(g - a)}$$

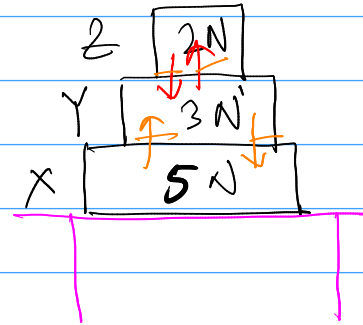
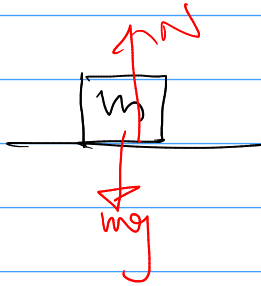
$$\boxed{a = g}$$

$$\boxed{N = 0}$$



Law III:

Action = - Reaction



$$F_{ZY} = 2N$$

$$F_{YT} = 2N$$

$F_{12} = \text{Force on 1 due to 2}$

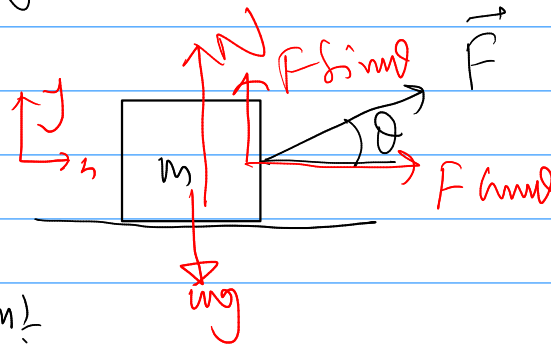
$$F_{TX} = 10N$$

T

$$F_{XY} = 8N$$

$\overline{F_{mty}}?$

$$F_{XT} = 10N$$



$a_z?$

Free body diagram:

$$\sum F_y = 0$$

$$\Rightarrow N + F \sin \theta - mg = 0$$

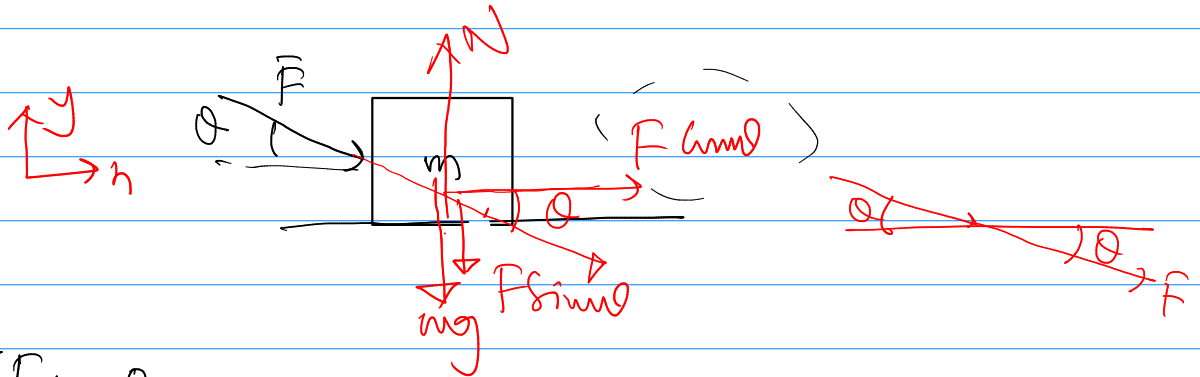
$$\Rightarrow N = mg - F \sin \theta$$

Apparent weight

$$\sum F_x = ma$$

$$\Rightarrow F_{\text{normal}} = ma$$

$$\Rightarrow \boxed{a = \frac{F_{\text{normal}}}{m}} \quad \checkmark \quad m/s \checkmark$$



$$\sum F_y = 0$$

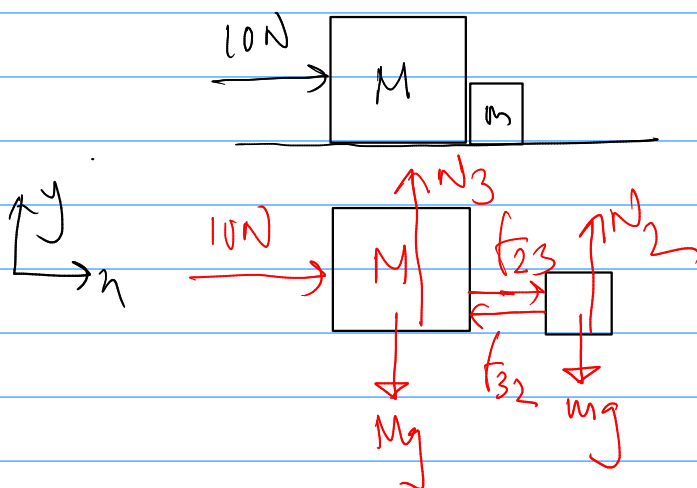
$$\Rightarrow N - mg - F \sin \theta = 0$$

$$\Rightarrow \boxed{N = mg + F \sin \theta}$$

$$\sum F_x = ma$$

$$\Rightarrow F_{\text{normal}} = ma$$

$$\Rightarrow \boxed{a = \frac{F_{\text{normal}}}{m}}$$



$$M = 3 \text{ kg}$$

$$m = 2 \text{ kg}$$

$$a = ?$$

$$\boxed{|f_{23}| = |f_{32}|}$$

2kg body:

$$\sum F_y = 0$$

$$\Rightarrow N_2 - mg = 0$$

$$\Rightarrow \boxed{N_2 = mg}$$

$$\sum F_x = ma$$

$$\Rightarrow \boxed{f_{23} = 2a} \quad \text{--- (1)}$$

3kg body:

$$\sum F_y = 0$$

$$N_3 - Mg = 0$$

$$\Rightarrow \boxed{N_3 = Mg}$$

$$\sum F_x = ma$$

$$\Rightarrow 10 = f_{32} = 3a$$

$$\Rightarrow \boxed{10 = f_{23} = 3a} \quad \text{--- (II)}$$

(I) + (II)

$$\cancel{f_{23}} = 2a$$

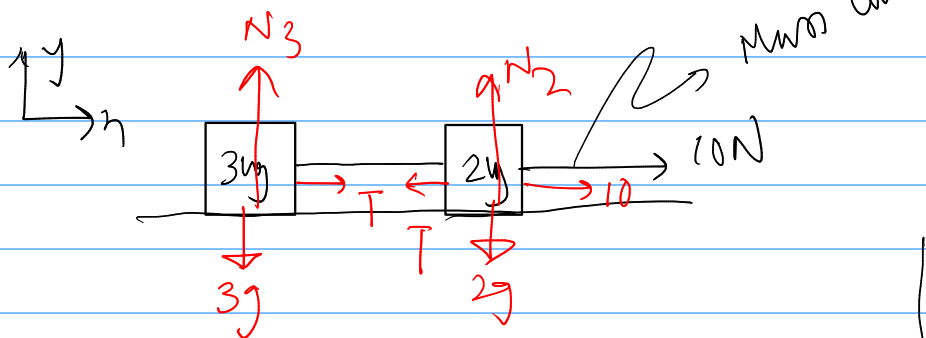
$$10 - \cancel{f_{23}} = 3a$$

$$10 = 5a$$

$$\Rightarrow \boxed{\vec{a} = 2 \text{ m/s}^2 \hat{i}}$$

$$(I) \Rightarrow f_{23} = ma = 2 \times 2 = 4 \text{ N}$$

$$10 - f_{23} = 10 - 4 = 6 \text{ N}$$



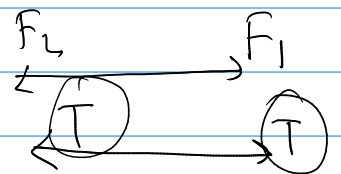
$$a = ?$$

3kg body:

$$\sum F_y = 0$$

$$\Rightarrow N_3 - 3g = 0$$

$$\Rightarrow \boxed{N_3 = 3g}$$



$$F_1 \neq F_2$$

$$F_{\text{net}} = F_1 - F_2$$

$$a \neq 0$$

$$a = \frac{F}{m} = \frac{F_{\text{net}}}{0}$$

$$\sum F_n = ma$$

$$T = 3a \quad \text{--- (i)}$$

For 2kg body:

$$\sum F_y = 0$$

$$N_2 - 2g = 0$$

$$N_2 = 2g$$

$$\sum F_n = ma$$

$$10 - T = 2a \quad \text{--- (ii)}$$

$$\text{[(i) + (ii)]} \quad T = 3a$$

$$10 - T = 2a$$

$$10 = 5a$$

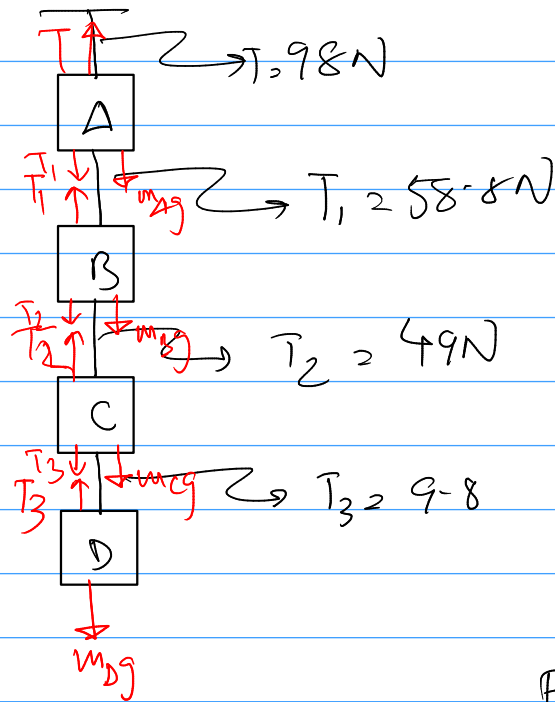
$$a = 2 \text{ m/s}^2$$

$$\text{(i) } T = 3a$$

$$T = 3 \times 2 = 6 \text{ N}$$

For 4kg body

$$18 - T = 4a$$



$$m_A = ?$$

$$m_B = ?$$

$$m_C = ?$$

$$m_D = ?$$

For D:

$$\sum F_y = 0$$

$$T_3 - m_D g = 0$$

$$\Rightarrow m_D = \frac{T_3}{g} = \frac{9.8}{9.8} = 1 \text{ kg}$$

For C:

$$\sum F_y = 0$$

$$T_2 - T_3 - m_C g = 0$$

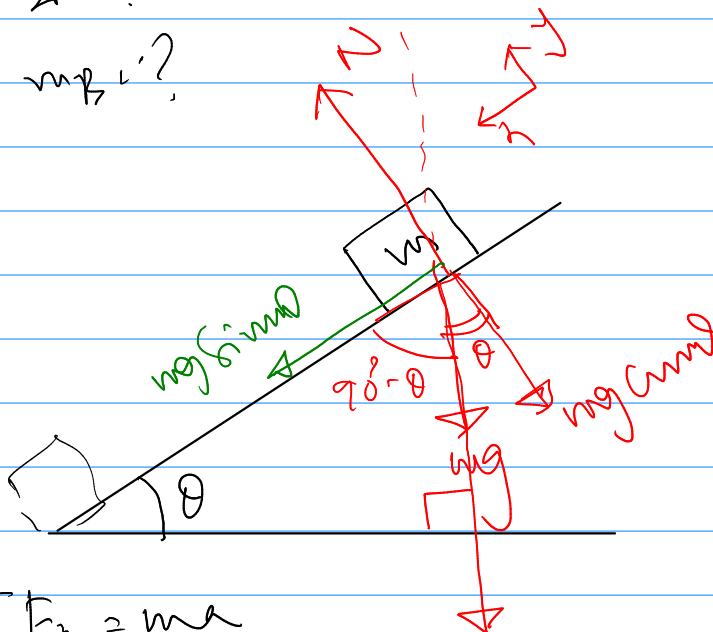
$$\Rightarrow m_C = \frac{T_2 - T_3}{g}$$

$$\Rightarrow m_C = \frac{49 - 9.8}{9.8}$$

$$\Rightarrow \boxed{m_C = 4 \text{ kg}}$$

At ω
 $m_A = ?$

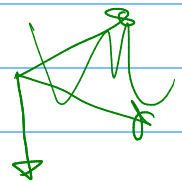
$m_B = ?$



$$\sum F_x = ma$$

$$\Rightarrow mg \sin \theta = ma$$

$$\Rightarrow \boxed{a = g \sin \theta}$$



$$\sum F_y = 0$$

$$\Rightarrow N - mg \cos \theta = 0$$

$$\Rightarrow \boxed{N = mg \cos \theta}$$

Apparent weight